

Consumer Behavior Analytics

Business Problem Statement & Analyst Project Brief

Retail Intelligence Initiative

1. The Business Challenge

What is the problem?

Our retail business is experiencing shifts in how customers shop — and we are not yet fully equipped to act on those changes. Purchasing patterns are evolving across age groups, product categories, and shopping channels (in-store vs. online). Without a clear picture of what is driving these shifts, we risk misallocating marketing budgets, missing revenue opportunities, and losing customer loyalty to more data-savvy competitors.

Core Business Question

"How can we use our customer shopping data to identify meaningful trends, strengthen customer engagement, and sharpen our marketing and product strategies?"

2. Why This Matters

Understanding customer behavior is no longer a nice-to-have — it is a competitive necessity. Here is what is at stake:

Business Area	What We Stand to Gain
Revenue Growth	Identify which products, promotions, and channels generate the highest returns so we can invest more strategically.
Customer Retention	Understand what keeps customers coming back — and what causes them to leave — so we can act before we lose them.
Marketing Efficiency	Stop spending on campaigns that do not resonate. Target the right customers with the right message at the right time.
Inventory & Product Strategy	Align stock levels and product offerings with actual demand patterns, reducing waste and stockouts.
Competitive Advantage	Companies that act on data insights respond faster to market changes and outperform those that rely on intuition alone.

3. Key Questions This Project Will Answer

This analysis is designed to answer specific, actionable business questions:

- Which customer segments generate the most revenue — and which are at risk of churning?
- Do discounts and promotions actually drive repeat purchases, or do they attract one-time bargain hunters?
- How do seasonal patterns affect buying behavior across product categories?
- Which payment methods and shopping channels (online vs. in-store) are growing, and which are declining?
- What role do product reviews and ratings play in influencing purchasing decisions?
- Which customer groups respond best to targeted marketing efforts?

4. Project Scope & Deliverables

This project spans four interconnected workstreams, each building on the last to produce a complete, decision-ready set of outputs.

#	Deliverable	What It Does	Who Benefits
1	Data Preparation & Cleaning (Python)	Transforms raw, messy customer data into a clean, analysis-ready format — removing errors, filling gaps, and structuring records consistently.	Analysts, Data Engineers
2	Data Analysis & Querying (SQL)	Organises the cleaned data into a structured database and runs queries to surface customer segments, loyalty patterns, and key purchase drivers.	Analysts, Business Managers
3	Interactive Dashboard (Power BI)	Presents findings in a visual, filterable dashboard that allows non-technical stakeholders to explore trends and patterns without touching any data.	Executives, Marketing, Sales Teams
4	Report & Presentation	Delivers a written summary of findings and a slide deck with clear business recommendations — actionable insights, not just charts.	All Stakeholders
5	GitHub Repository	Stores all code, queries, and files in an organised, version-	Technical Teams, Future Analysts

		controlled repository — ensuring the work is reproducible, auditable, and shareable.	
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5. Analyst Approach & Methodology

As the lead analyst on this project, the following methodology will be applied to ensure findings are both statistically sound and practically useful:

Step 1 — Data Audit & Preparation

Before any analysis begins, the raw dataset will be thoroughly reviewed. This includes identifying and resolving missing values, correcting inconsistent data formats (e.g., date fields, category labels), removing duplicate records, and engineering new features — such as customer tenure, average basket size, and purchase frequency — that are critical for segmentation and behavioural analysis. Tool: Python (pandas, numpy).

Step 2 — Structured Data Modelling

The cleaned dataset will be loaded into a relational database. SQL queries will then be used to aggregate data by customer segment, product category, season, and channel. This step produces the summary tables that feed directly into the dashboard and report. Tool: SQL (PostgreSQL or equivalent).

Step 3 — Insight Generation & Validation

Statistical and descriptive analysis will be used to identify trends, correlations, and anomalies. Key metrics — such as customer lifetime value, repeat purchase rate, discount elasticity, and channel conversion rates — will be calculated and validated against business expectations before being surfaced.

Step 4 — Visualisation & Communication

Findings will be translated into an interactive Power BI dashboard with filters for region, product category, time period, and customer segment. A written report and executive presentation will accompany the dashboard, each tailored to its audience — technical detail in the report, clear narrative in the slides.

6. Expected Outcomes & Business Value

Upon completion, this project will equip the business with:

- A clear, evidence-based view of who our most valuable customers are and what drives their loyalty.
- Actionable segmentation that marketing and sales teams can use to personalise campaigns immediately.
- An understanding of which promotions and discounts actually improve long-term revenue — not just short-term volume.

- Seasonal demand intelligence to guide inventory planning and promotional scheduling.
- A reusable, automated dashboard that stakeholders can interrogate independently — reducing reliance on ad hoc data requests.
- A version-controlled codebase that future analysts can build upon, maintain, and extend.

7. A Note on Audience

This project is designed to serve two distinct audiences simultaneously:

For Business Stakeholders	For Technical Practitioners
The report, dashboard, and presentation are written in plain language. No prior data or technical knowledge is required to understand the findings or act on the recommendations.	All Python scripts, SQL queries, and data transformation steps are fully documented in the GitHub repository, with inline comments and a clear README to enable replication and extension.